

O’Neill School of Public and Environmental Affairs

The Educational Effects of COVID-19 Vaccine Eligibility

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SECTION 1

Introduction & Background

Paper Overview

- Research Question: What is the effect of COVID-19 Vaccine eligibility on students' learning outcomes?
  - In this paper, I examine the effect of being eligible for COVID-19 vaccine (Pfizer) on ILEARN test scores (Indiana's statewide standardized assessments)
- Data from Indiana Department of Education
- Being eligible for COVID-19 vaccine → reduced attendance → reduced test scores

Introduction

- COVID-19 pandemic: unprecedented disruptions to K-12 education in the US
  - School closures, quarantines, shifts to remote instruction
- Large literature on learning losses from the pandemic
  - the effects of school closures and remote instruction (Betthauser et al 2023)
- No study directly examines the educational effects of vaccine rollout

Background

- On May 12, 2021, Indiana opened Pfizer-BioNTech vaccination to children aged 12-15
  - Students born on or before May 12, 2009 = eligible immediately
  - Students born after November 3, 2009 = not eligible until November 3, 2021
    - six-month difference in vaccine eligibility
- In IDOE data, I create two birth cohorts
  - Early Eligible Cohort:
    - Students who were born between May 12, 2008 and May 10, 2009
  - Late Eligible Cohort:
    - Students who were born between Nov 3, 2009 and Nov 3, 2010

All K-12 public and non-public school closures

March 19, 2020

Schools were allowed to begin reopening (with mask)

July 1, 2020



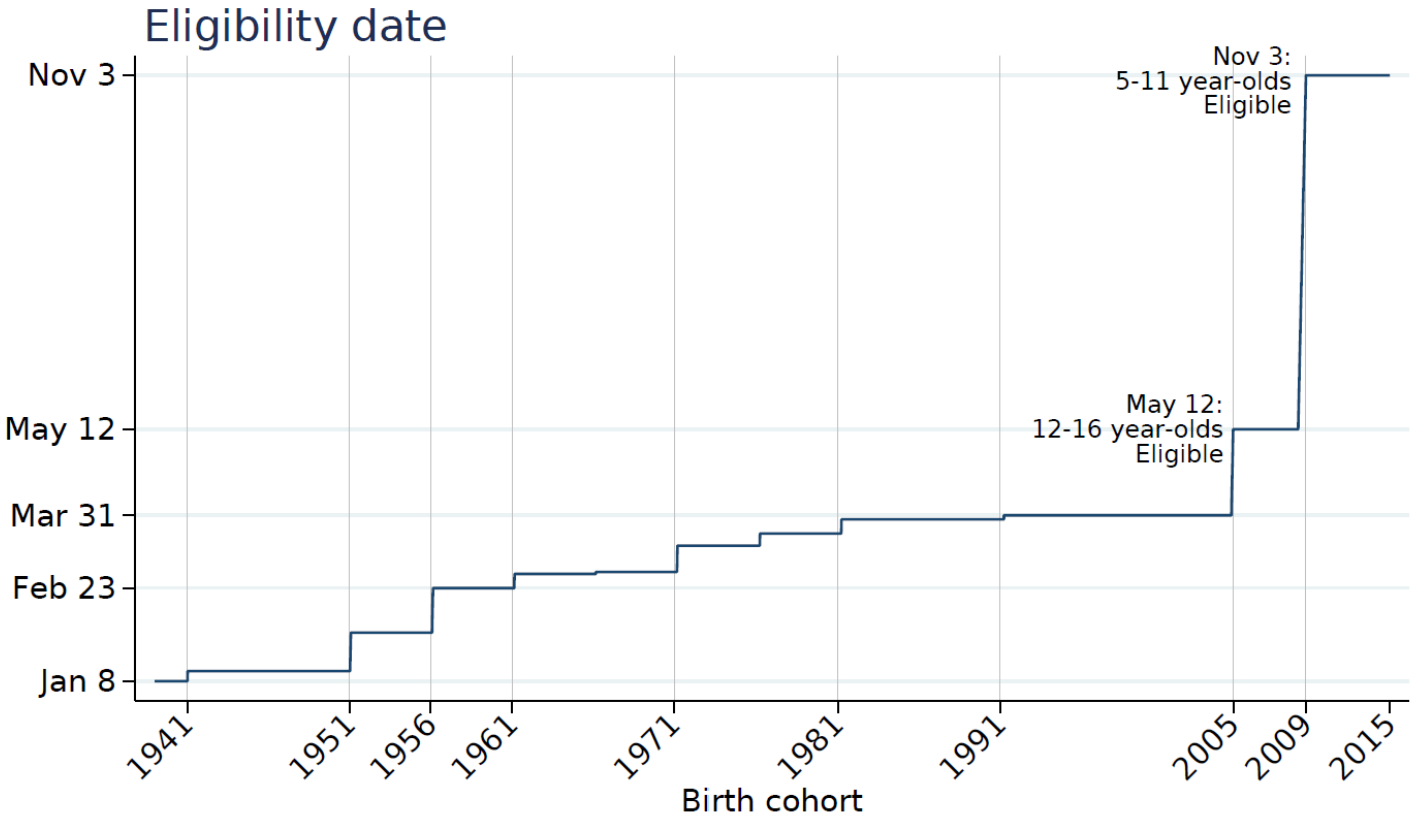
Remote learning

March, 2021

By spring 2021, nearly all Indiana students were back in person

May 12, 2021

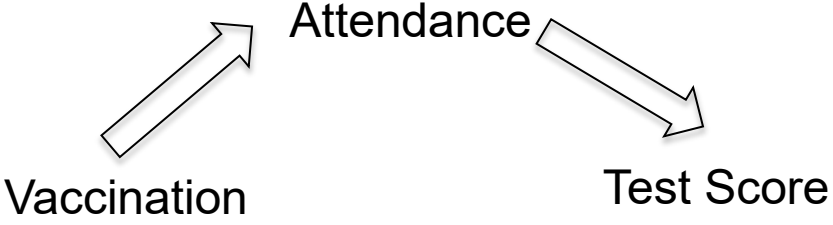
Indiana opened Pfizer-BioNTech vaccinations to all children aged 12-15.



Notes: Figure shows the date (in 2021) Indiana expanded eligibility for the COVID-19 vaccines to different age groups, by day of birth.

Hypothesis

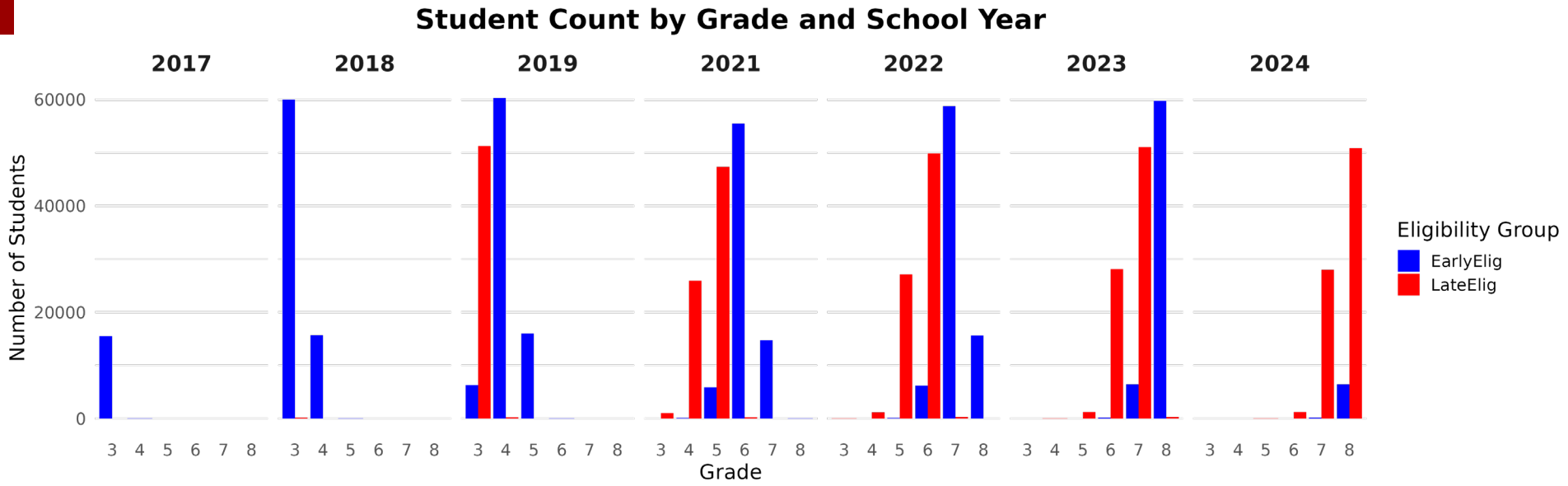
- Vaccination may
  - Increase attendance: Exempt from quarantine rules, reduced risk of missing school due to infection, families feel safer sending children to school, etc
    - → higher test scores: more days in school
  - Decrease attendance: post-vaccine side effects (both short run and long run absences)
    - → lower test scores: less days in school

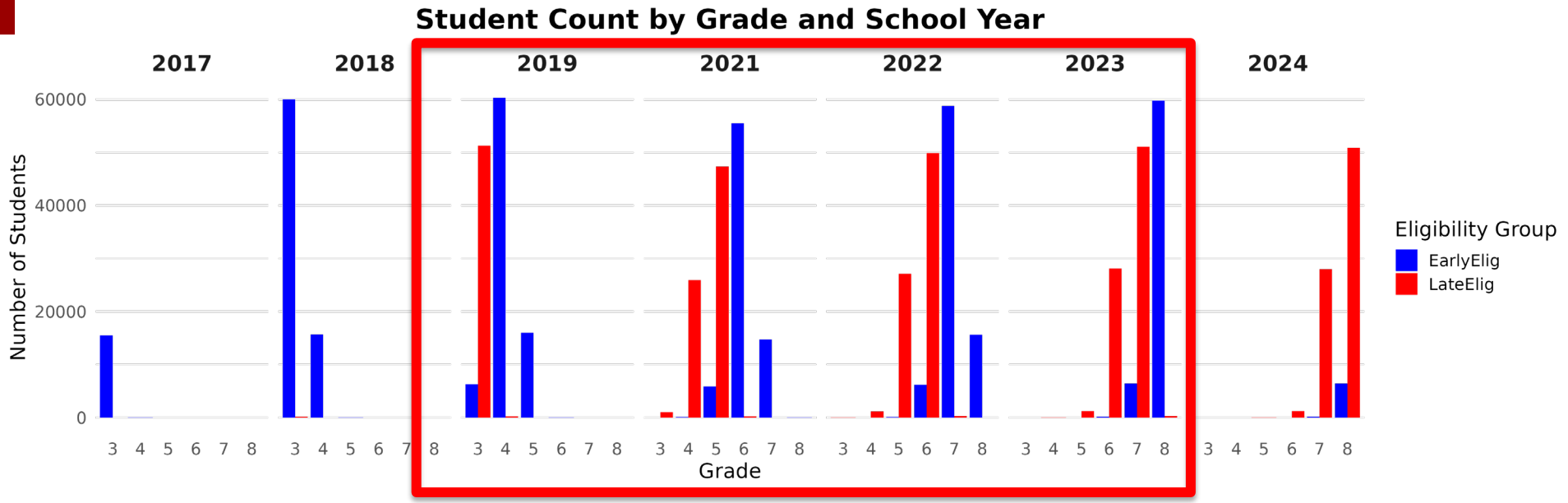


Background

- ILEARN (Indiana Learning Evaluation and Readiness Network)
  - Indiana's standardized assessment for students in grades 3 – 8
  - Subjects tested:
    - ELA (Grades 3-8)
    - Mathematics (Grades 3-8)
    - Science (Grades 4 & 6)
    - Social Studies (Grade 5)
    - Biology (high school. Usually Grade 10)
  - Administered every spring (around late April through Mid-May)

Subject	Grade	Approaching Proficiency	At Proficiency	Above Proficiency	LOSS	HOSS
English/ Language Arts	3	5416	5460	5515	5060	5760
	4	5444	5493	5547	5090	5810
	5	5472	5524	5595	5110	5850
	6	5492	5544	5604	5130	5870
	7	5507	5568	5629	5130	5890
	8	5511	5577	5638	5150	5920
Mathematics	3	6382	6425	6488	6080	6730
	4	6429	6474	6541	6100	6800
	5	6453	6510	6566	6110	6850
	6	6488	6545	6605	6110	6870
	7	6493	6562	6625	6120	6920
	8	6509	6590	6651	6120	6950
Science	4	145	163	180	100	220
	6	347	363	372	300	420
	Biology	546	561	573	500	620
Social Studies	5	8477	8502	8543	8350	8650


Student Count by Grade and School Year
2017
2018
2019
2021
2022
2023
2024
60000
40000
20000
0
Number of Students
3 4 5 6 7 8
Grade
Eligibility Group
EarlyElig
LateElig


Student Count by Grade and School Year
2017
2018
2019
2021
2022
2023
2024
60000
40000
20000
0
Number of Students
3 4 5 6 7 8
Grade
Eligibility Group
EarlyElig
LateElig

SECTION 2

Data and Methodology

Data

- Data from Indiana Department of Education
  - Attendance (days attended, days absent excused, days absent unexcused, etc)
  - ILEARN test scores
  - Demographic information (ethnicity, gender, birth date, etc)
- Longitudinal dataset (Repeated Measure)
  - For each unique Student ID:
    - days attended for each year (or grade)
    - ILEARN test scores for each year (from the time when the student was Grade 3 to Grade 8)

Method

$$TestScore_{igt} = \beta_{DID} \cdot \text{Early Eligible}_{igt} + a_i + b_t + \varepsilon_{igt}$$

- $TestScore_{igt}$: ILEARN test score for student “i” in birth cohort “g” in year “t”
- $Early\ Eligible_{igt}$: Dummy variable for being in the Early Eligible Group and after 2021
- a_i: Student Fixed Effects
  - Captures the fixed differences across students
- b_t: Year Fixed Effects
  - Captures the fixed differences across years
- Clustered at Student Level

Method

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Method

$$Y_{igt} = \sum_{h=2019; h \neq -2021}^{2023} [\theta_h(EarlyEligible_{ig} \times 1[e = h])] + a_i + b_t + \epsilon_{igt}$$

- Event Study Graph
  - Summation term is a set of interactions between a dummy for being in a early eligible group and each time period relative to treatment (2021)
  - Reference Year = 2021
    - Excluded term
- Implies that the θ_h terms trace out each of the simple DID's between period 2021 and period 2021 + e

Method

- Identifying assumptions for causal inference
  - Common Trend Assumption
    - Early Eligible Group and Late Eligible Group were following a similar trend during pre-treatment period, and the common trend continued to exist during the post-treatment period
  - No Anticipation Assumption
    - No behavioral changes in anticipation of COVID-19 eligibility

Method

$$y_{it} = \beta_0 + \beta_1 \,\mathrm{treat}_{it} + \sum_{t=2019}^{2023} \gamma_t \,\mathbf{1}\{\mathrm{year}_{it} = t\} + u_i + \varepsilon_{it},$$

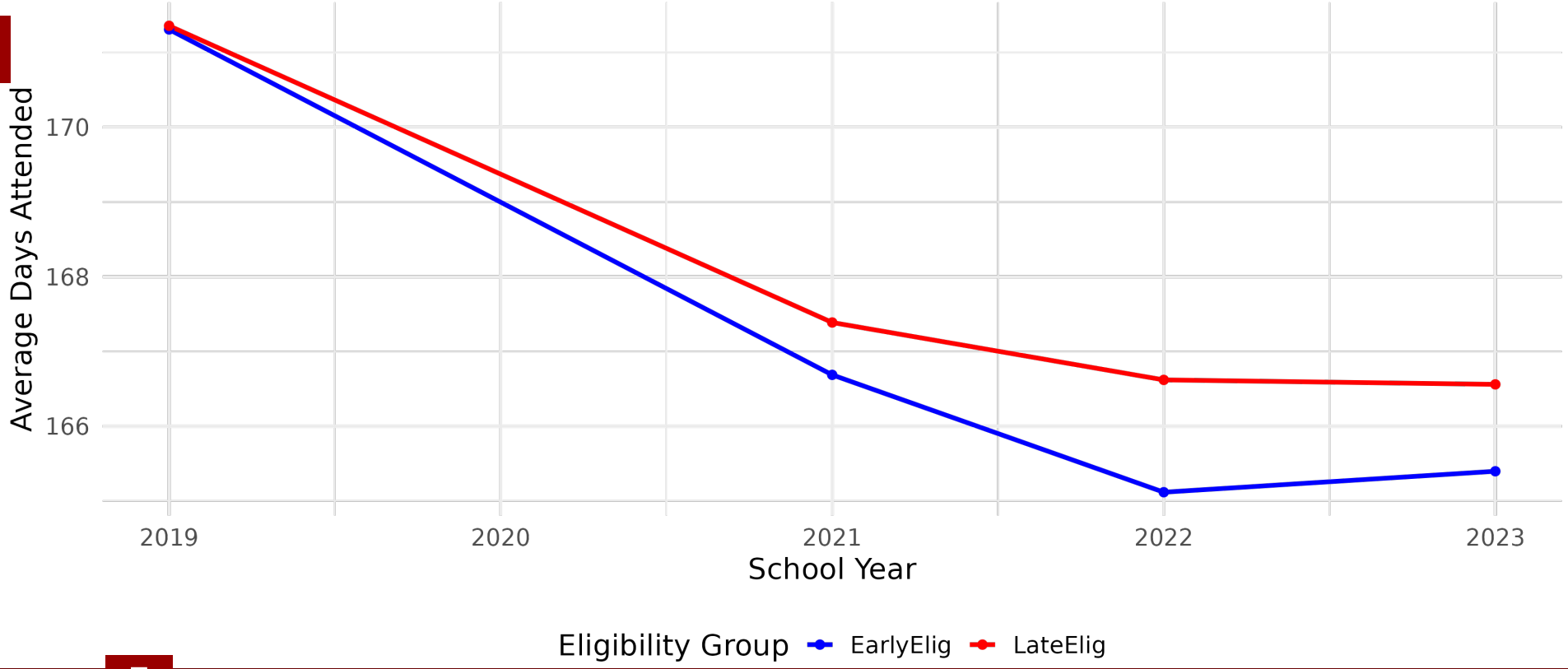
$$u_i \sim \mathcal{N}(0, \tau^2), \qquad \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2), \qquad u_i \perp \varepsilon_{it}.$$

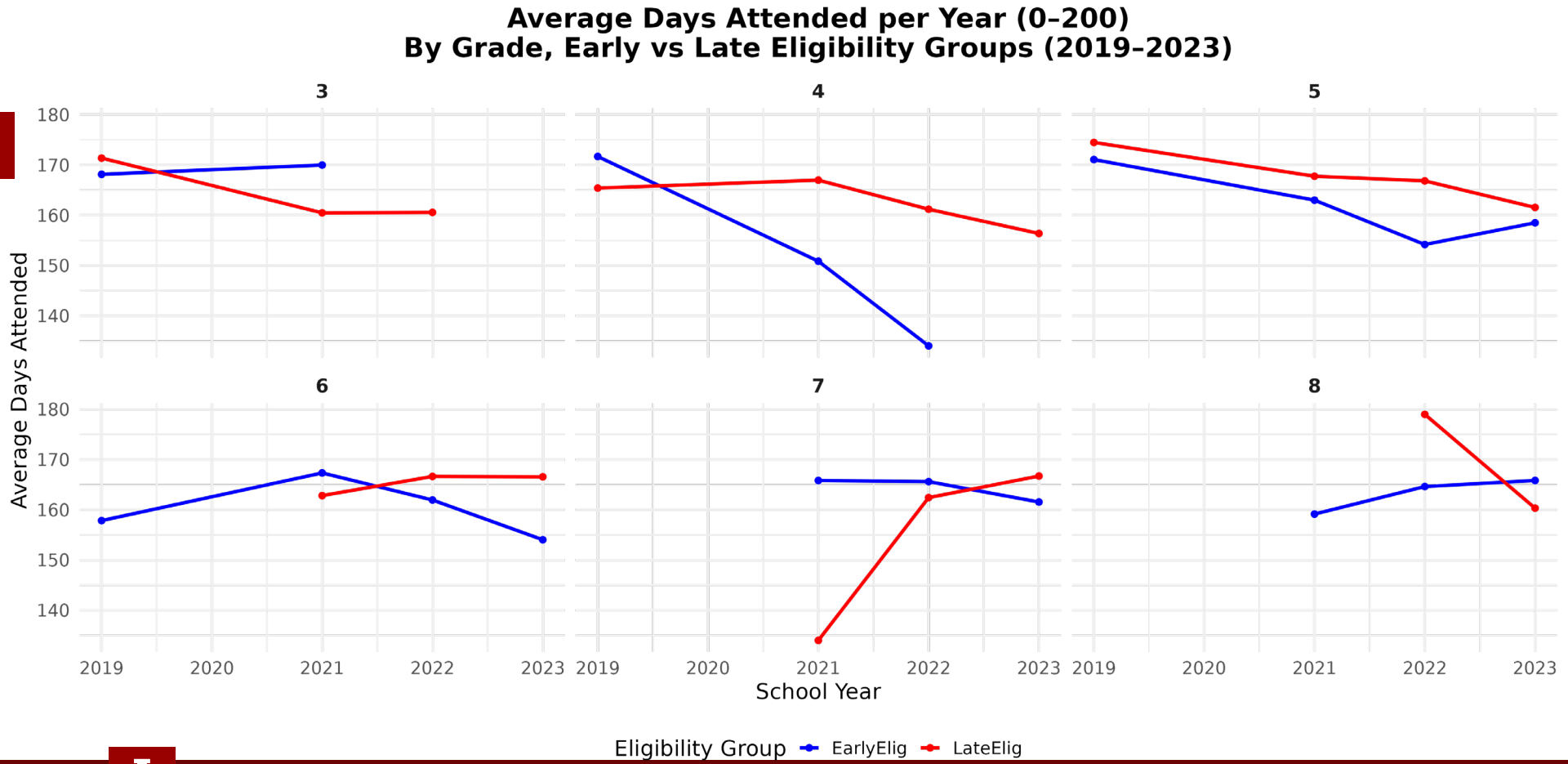
- Linear Mixed Effects Model with
  - γ_t: A fixed effect for each year
  - u_i: A random intercept for each student
  - ϵ_{it}: residual error

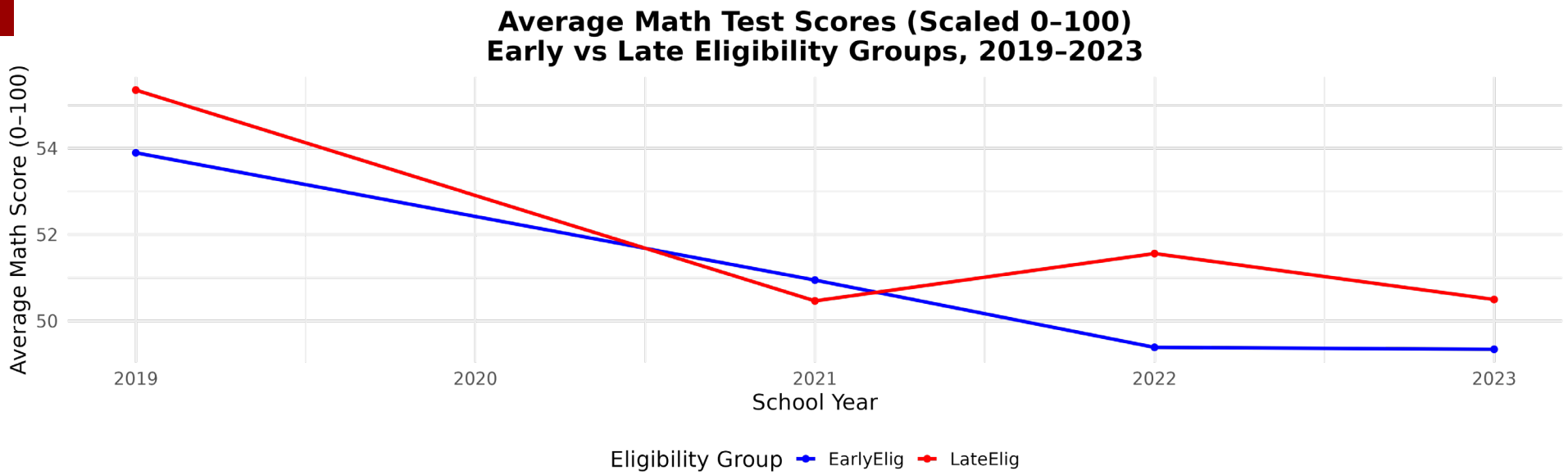
SECTION 3

Preliminary Graphs

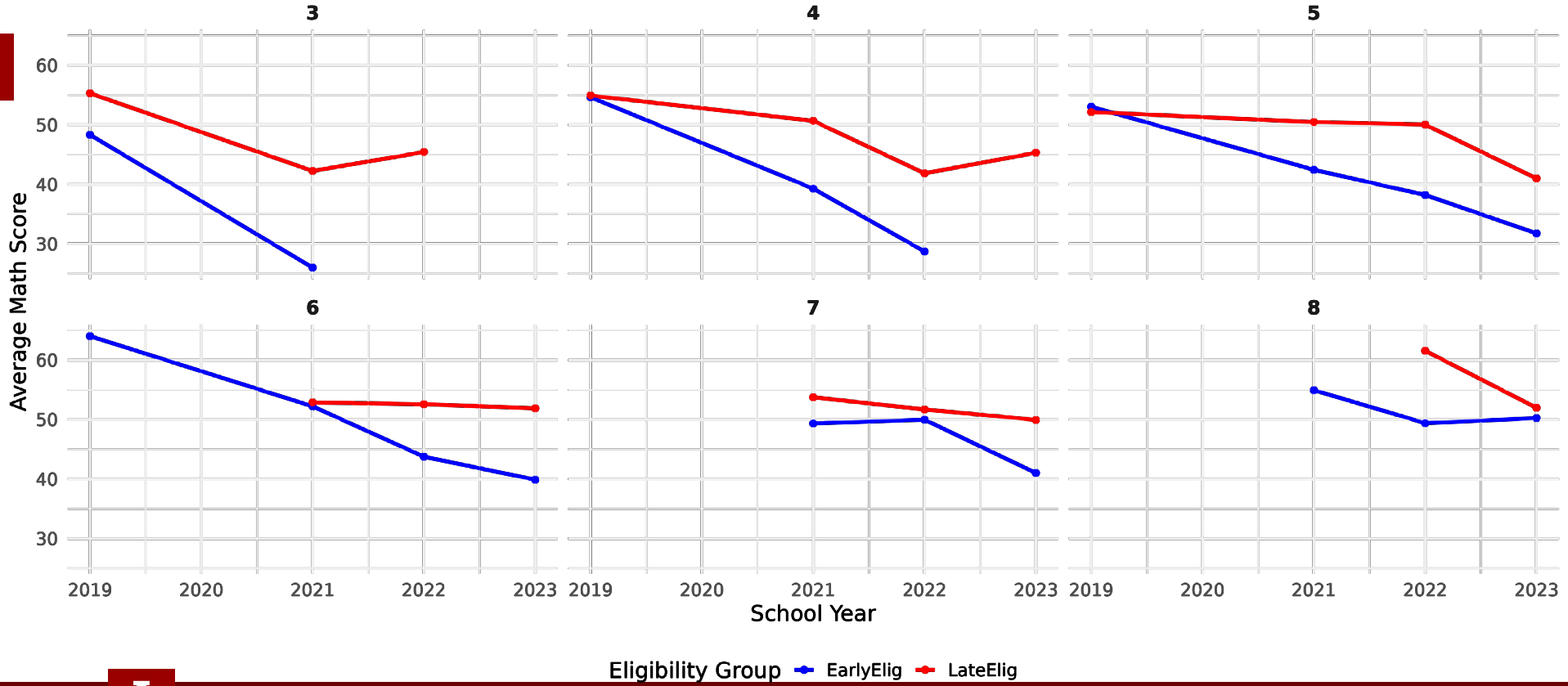
Average Days Attended per Year
Early vs Late Eligibility Groups (2019–2023)



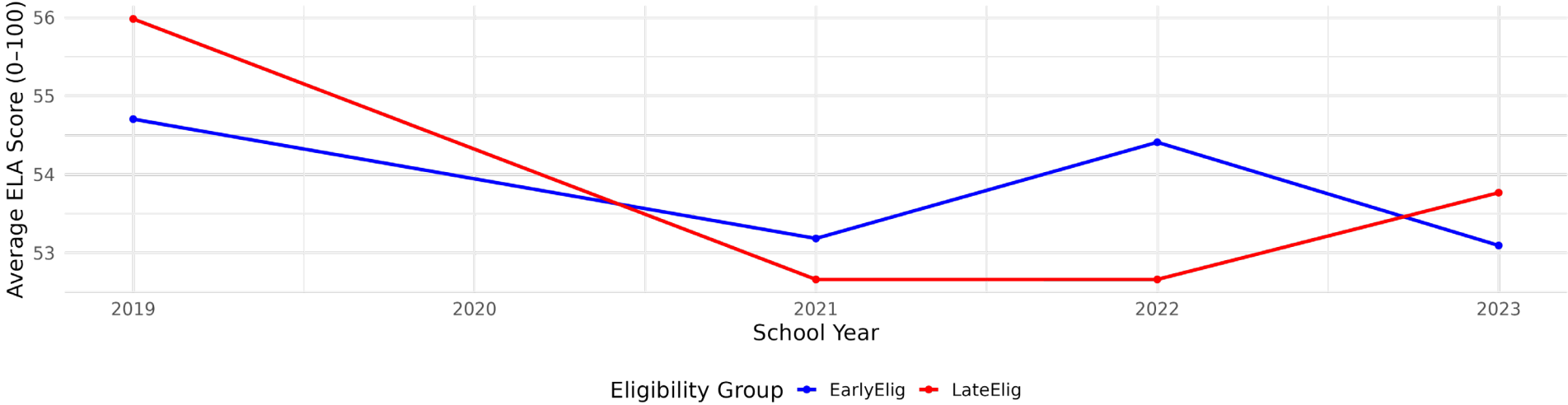

Average Days Attended per Year (0–200)
By Grade, Early vs Late Eligibility Groups (2019–2023)
3
4
5
6
7
8
180
170
160
150
140
Average Days Attended
2019
2020
2021
2022
2023
School Year
Eligibility Group
EarlyElig
LateElig


Average Math Test Scores (Scaled 0–100)
Early vs Late Eligibility Groups, 2019–2023
Average Math Score (0–100)
54
52
50
2019
2020
2021
2022
2023
School Year
Eligibility Group
EarlyElig
LateElig

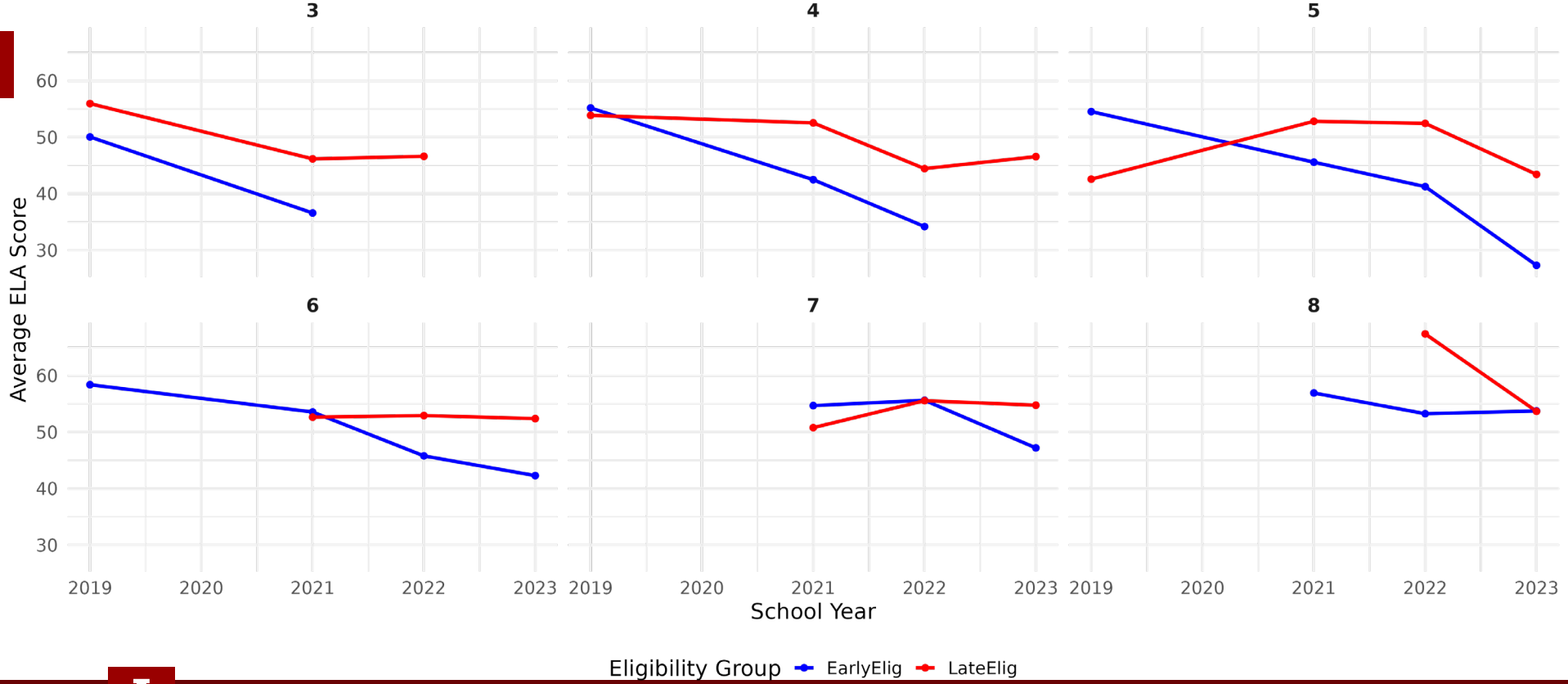
Average Math Test Scores (0–100)
By Grade, Early vs Late Eligibility Groups (2019–2023)



Average ELA Test Scores (Scaled 0–100)
Early vs Late Eligibility Groups, 2019–2023



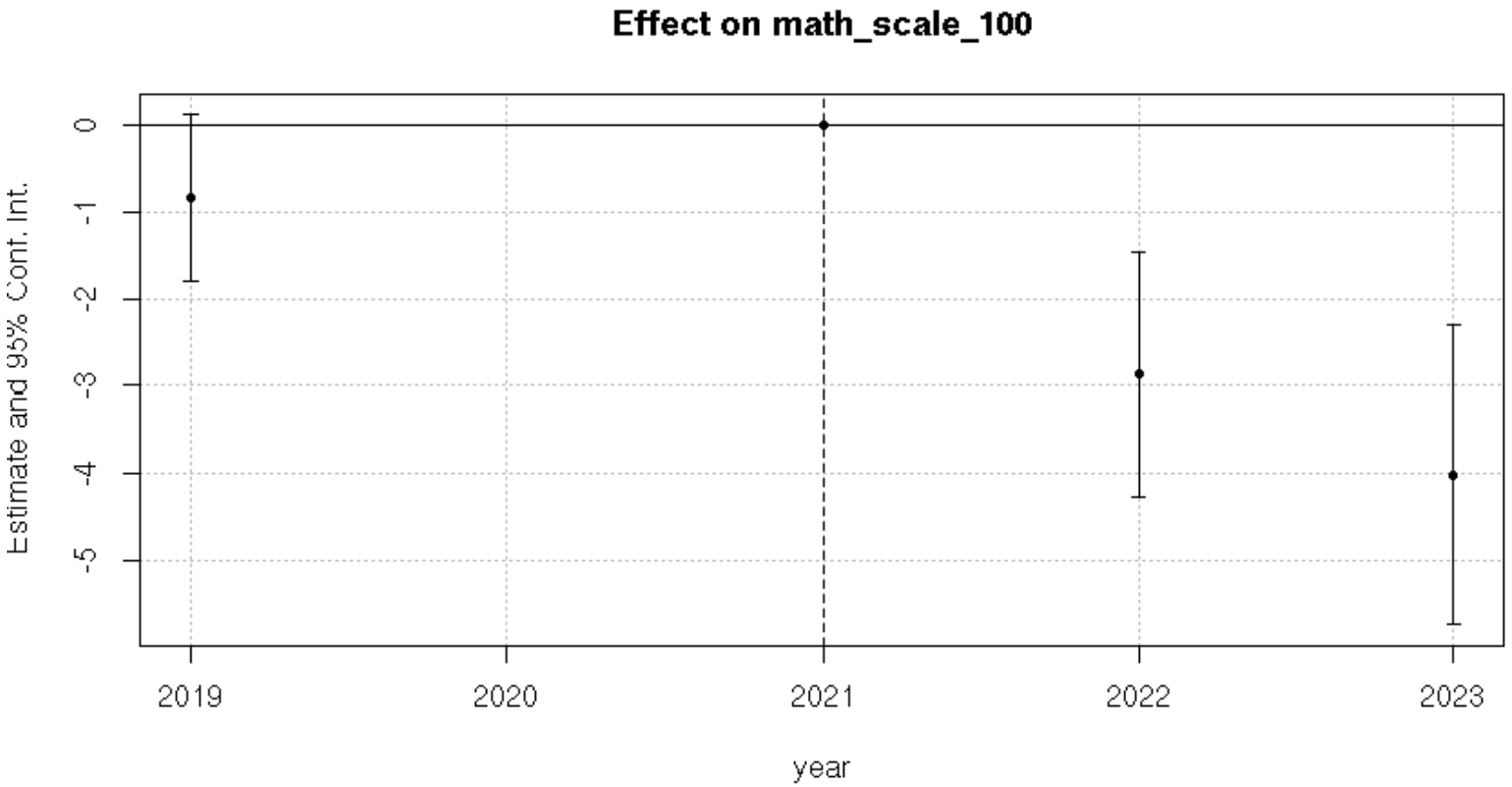
Average ELA Test Scores (0–100)
By Grade, Early vs Late Eligibility (2019–2023)

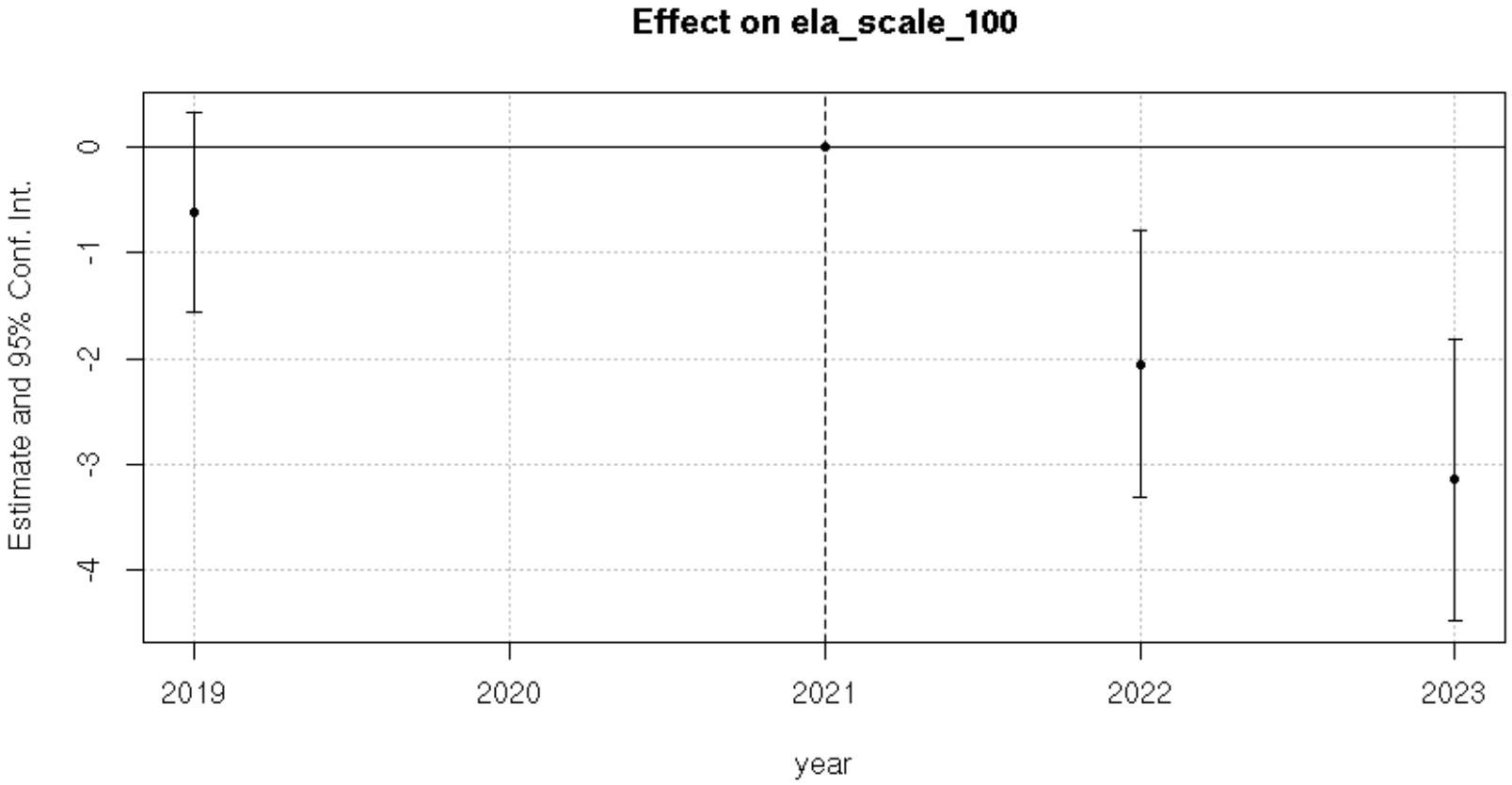


SECTION 4

Main Results

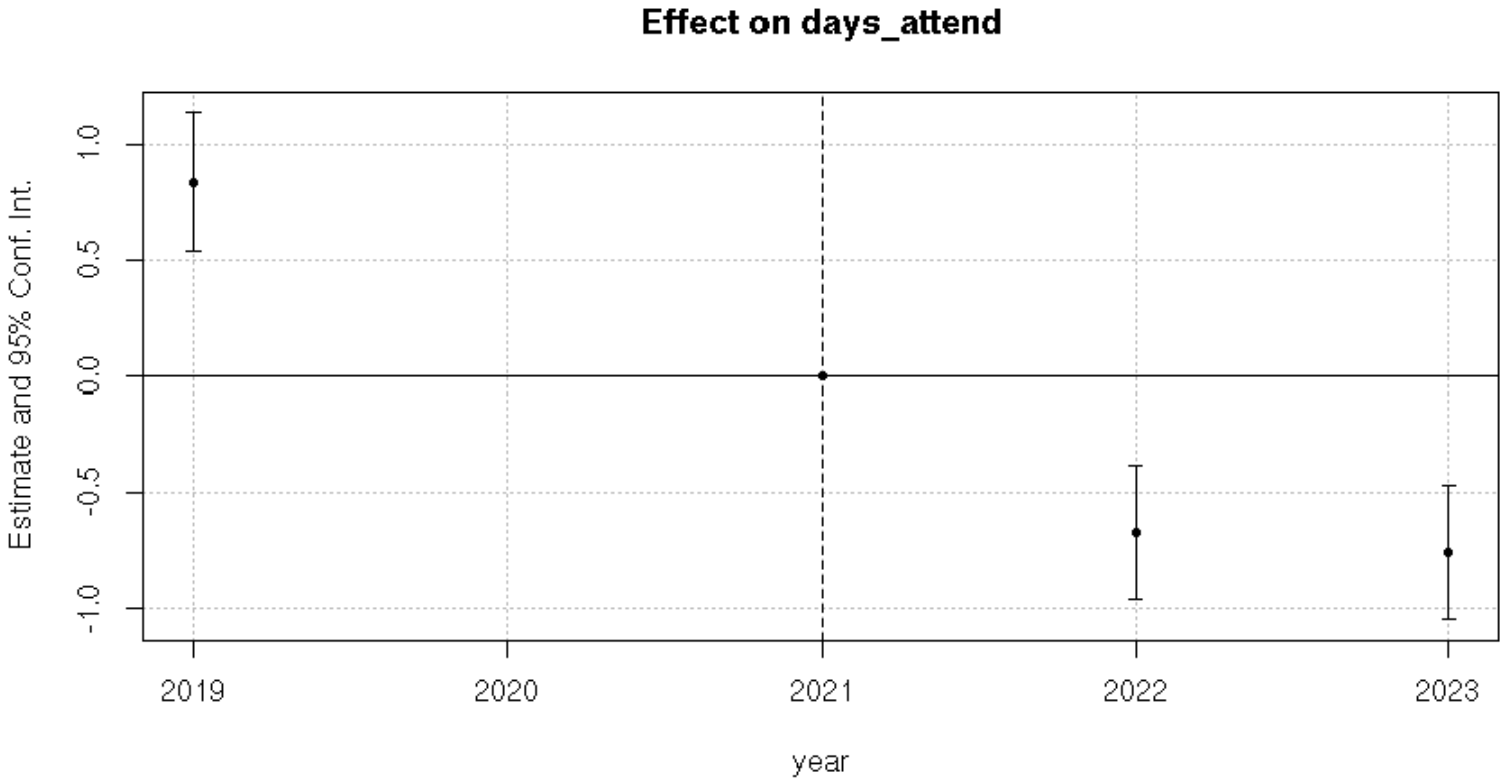
Fixed Effects Model


Effect on math_scale_100
Estimate and 95% Cont. Int.
0
-1
-2
-3
-4
-5
2019
2020
2021
2022
2023
year


Effect on ela_scale_100
Estimate and 95% Conf. Int.
0
-1
-2
-3
-4
2019
2020
2021
2022
2023
year

	M1 Math	M1 ELA	M2 Math	M2 ELA	M3 Math	M3 ELA	M4 Math	M4 ELA
treat_igt	0.256***	1.390***			−2.830***	−2.254***		
	(0.037)	(0.037)			(0.693)	(0.567)		
year = 2019 × treat_ig			−1.746***	−1.298***			−0.837+	−0.620
			(0.039)	(0.041)			(0.489)	(0.479)
year = 2022 × treat_ig			−2.653***	1.274***			−2.861***	−2.056**
			(0.033)	(0.036)			(0.717)	(0.645)
year = 2023 × treat_ig			−1.821***	−1.163***			−4.020***	−3.148***
			(0.039)	(0.038)			(0.881)	(0.682)
math_2018					0.809***		0.809***	
					(0.002)		(0.002)	
ela_2018						0.967***		0.967***
						(0.003)		(0.003)
Num.Obs.	580 213	580 213	580 213	580 213	262 836	262 836	262 836	262 836
R2	0.882	0.843	0.883	0.844	0.621	0.569	0.621	0.569
R2 Adj.	0.834	0.779	0.836	0.781	0.621	0.569	0.621	0.569
R2 Within	0.000	0.004	0.013	0.014	0.614	0.568	0.614	0.568
R2 Within Adj.	0.000	0.004	0.013	0.014	0.614	0.568	0.614	0.568
Std.Errors	by: ID	by: ID	by: ID	by: ID	by: ID	by: ID	by: ID	by: ID
FE: year	X	X	X	X	X	X	X	X
FE: ID	X	X	X	X				

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001


Effect on days_attend
Estimate and 95% Conf. Int.
1.0
0.5
0.0
-0.5
-1.0
2019
2020
2021
2022
2023
year

Conclusion

- After being eligible for COVID-19 vaccine, students aged 12 – 16 got less ILEARN scores
  - Probably due to less attendance rate
- Implications for policy makers
  - Understanding why attendance fell and how schools responded will be important for designing future public-health policies

Thank you!



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